

Jay Fan

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WORK EXPERIENCE

Software Engineer

2025 – Present

Rilla

- Designed and implemented the core **Stripe**-based payment infrastructure to support in-product billing along with frontend web and mobile components to enable self-serve payments. Processed payments totalling **over \$2 million** to date.
- Implemented robust fallback and dispatch policies for the livestreaming orchestration service in **Python**. **Reduced** the number of livestream crashes and failures by **95%**.
- Conducted **A/B testing** on the livestreaming feature to increase the week over week retention. Improved the week over week retention from **12% to over 40%**.

Software Engineer

2024 – 2025

Relay

- Led the design and technical discussions of a new integration system that would allow users to seamlessly integrate with 3rd party services (e.g. **Intuit Quickbooks**, **Xero**, **Gusto**, etc). Utilization of integration features **increased by 20%** compared to previously.
- Designed and implemented a new ingestion system using **AWS Lambda**, **S3**, and **dynamoDB** to enable users to directly upload bills onto our platform. This new workflow **increased the use of our bill pay feature by 50%**.
- Overhauled the onboarding flow of our expense management service using **React** and **TypeScript** to increase our conversion rate. The completion rate of the new onboarding flow was **45% higher** than the old design.
- Closely monitored the **datadog** logs in the alerts channel to keep a pulse on the health of our services. Tickets would be created and worked on for frequently repeating errors to continuously improve our system, reduce technical debt, and reduce noise. The frequency of alerts was **reduced by about 20%**.
- Collaborated with cross-functional teams and product designers to create an in-house frontend component library using **React** and **Tailwind**. All teams have begun the migration process to adopt the component library, while the work to create more complex components is ongoing.

Software Engineer

2022 – 2024

Laserfiche

- Migrated our **AWS EC2** containers to a new architecture utilizing **Kubernetes** services, **AWS SQS** queues, and **AWS SNS** topics to improve the user experience and scalability of our application. Users were able to get instant feedback upon making requests, and the new service has not dropped a single request since the migration.
- Enhanced the performance of two **C#** microservices that made requests to our **PostgreSQL** database by implementing batch and asynchronous **RESTful APIs**. Our biggest clients with more than 20,000 users observed a **runtime reduction of 93%** when performing large-scale operations.
- Created **RESTful APIs** using **TypeScript** and **Node** to serve client requests on our account control system. Daily unit, functionality, and webapi test results are monitored via **Grafana** dashboards, and they currently have a success rate of **98% over the last 90 days**.
- Implemented a user feedback system in **Python** and deployed via **AWS Lambda** functions. The system stored messages in **AWS Redshift** and **S3**, and allowed users to attach image and GIF files. After testing, the system was deployed to all user-facing products and is the primary way that we gather feedback from clients.
- Created an internal web application using **React** and **Flask**, enabling non-technical users to track their events in Redshift via drop-down menus. Managed the deployment with **Docker** containers, using **Nginx** as the reverse proxy for efficient application delivery.

EDUCATION

Master of Data Analytics

Western University · Ontario, Canada · 2020

Bachelor of Medical Science, Biochemistry of Infection and Immunity

Western University · Ontario, Canada · 2019

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, C#, SQL, Bash, C

Frameworks: React, Redux, Tailwind, Entity Framework, Vue, RxJS, Express, Angular, Flask, Django, Plotly

Tools & Technologies: Kubernetes, PostgreSQL, Node, Datadog, Docker, Cloud Formation, Redshift, S3, DynamoDB, LaunchDarkly, Nginx, Kibana, Grafana, Git